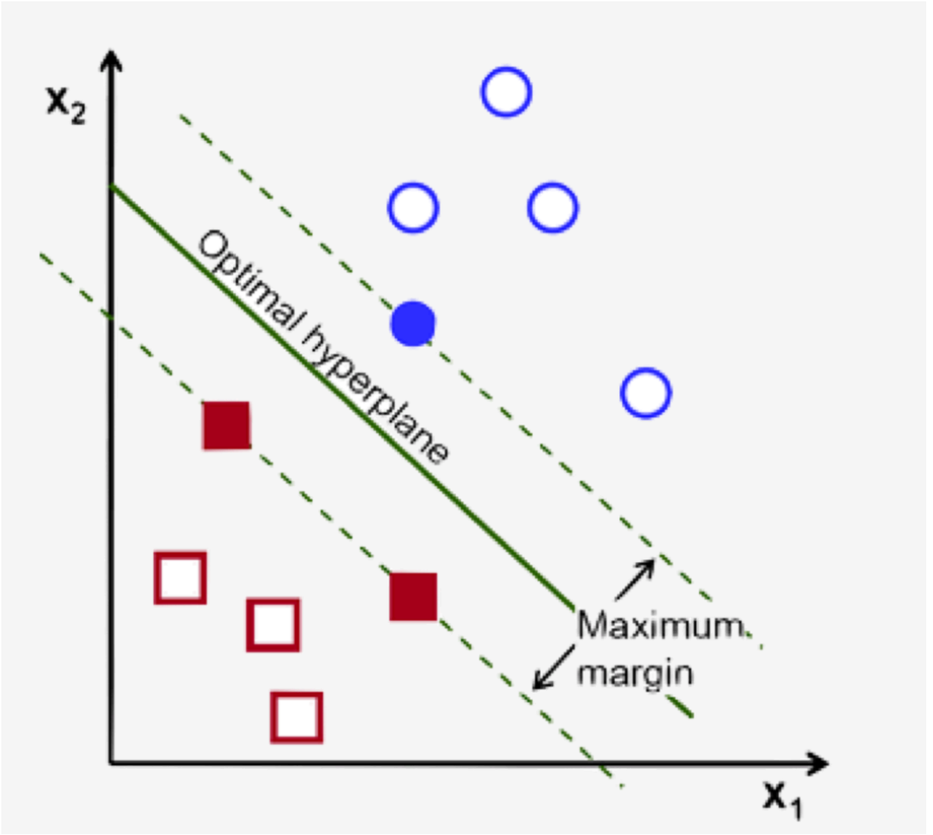
Support Vector Machines

CS172 · Computer Vision I

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From geometry to a learning objective

1. **Choose a boundary geometrically**

   Distance, scale, maximum margin, and support vectors.

2. **Optimize with the dual**

   Multipliers, constraints, and a small worked solution.

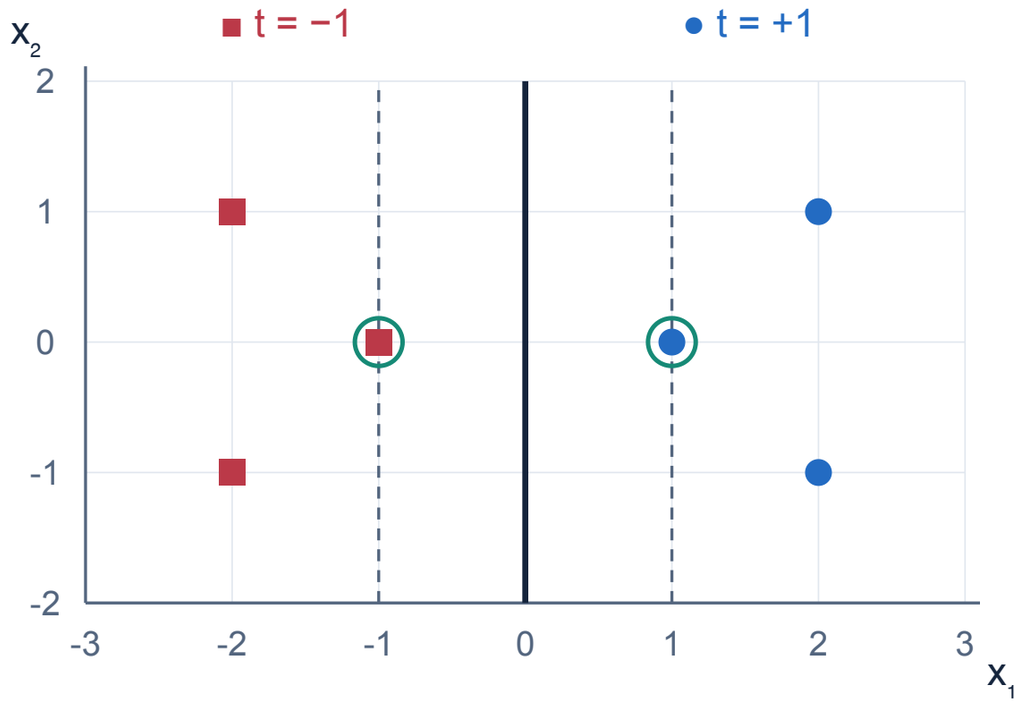
3. **Handle violations and nonlinear patterns**

   Slack variables, regularization, and the kernel trick.

4. **Extend to multiclass learning**

   Scores, hinge loss, and gradient-based optimization.

Start with two classes



Training data: (x_i, t_i), with

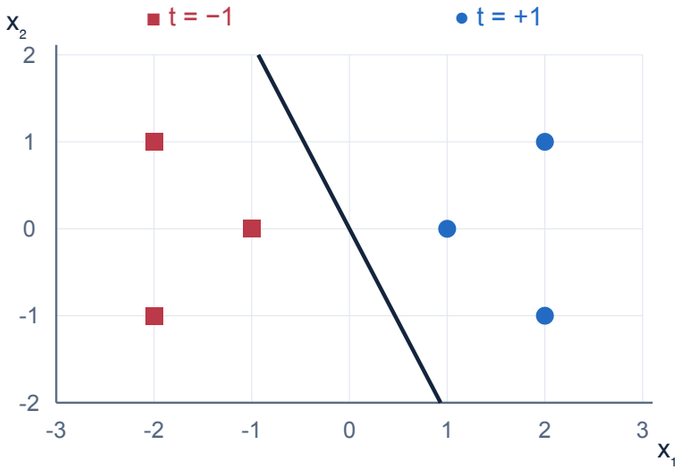
$$x_i \in \mathbb{R}^D, \qquad t_i \in \{-1, +1\}$$

$$f(x) = w^\top x + b$$

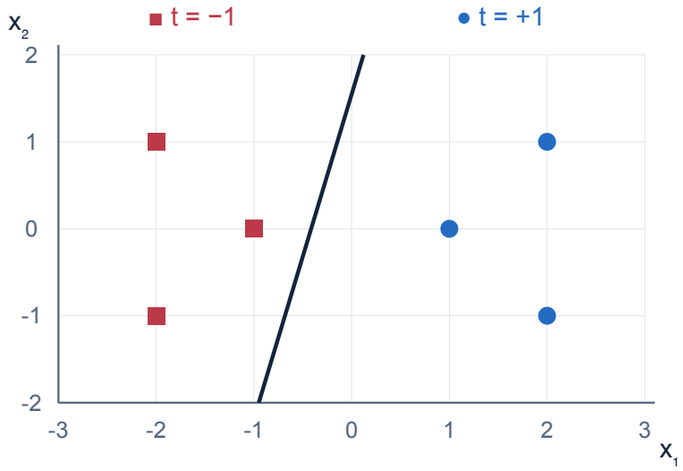
Predict the sign of $f(x)$.

A zero score is a tie. Scores are not probabilities.

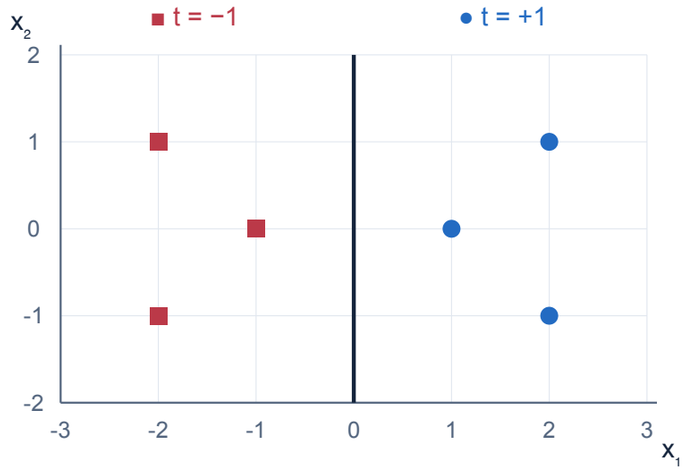
Correct classification leaves many choices



nearest signed distance = 0.91



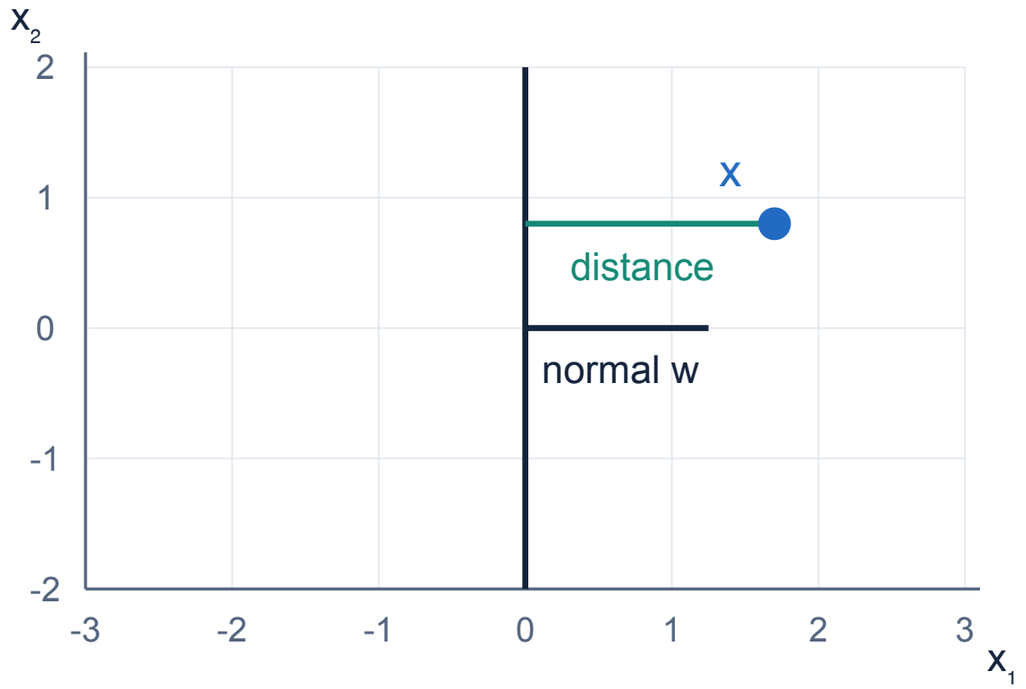
nearest signed distance = 0.57



nearest signed distance = 1.00

> Choose the separator with the largest distance to its nearest training example.

The weight vector is normal to the boundary



$$\mathcal{H} = \{x : w^\top x + b = 0\}$$

w controls orientation.
b shifts the boundary.

On the boundary, $w^\top(x + \delta) + b = w^\top \delta$.

> Moving perpendicular to w stays in the plane.

Normalize the score to get a distance

$$\text{signed distance to } \mathcal{H} = \frac{w^\top x + b}{\|w\|_2}$$

For labeled example i, the geometric margin is

$$\gamma_i = \frac{t_i(w^\top x_i + b)}{\|w\|_2}$$

> Positive means correct; larger means farther from the boundary on the correct side.

The dataset margin is $\gamma = \min_i \gamma_i$.

The score scale is arbitrary

$$(w, b) \quad \text{and} \quad (aw, ab), \quad a > 0$$

They define the same boundary and predictions.

Functional margin

$$t_i(w^\top x_i + b)$$

Multiplies by a.

Geometric margin

$$\frac{t_i(w^\top x_i + b)}{\|w\|_2}$$

Stays unchanged.

> An unnormalized score alone cannot identify the widest margin.

Fix the closest functional margin to one

For separable data, rescale a separator so that

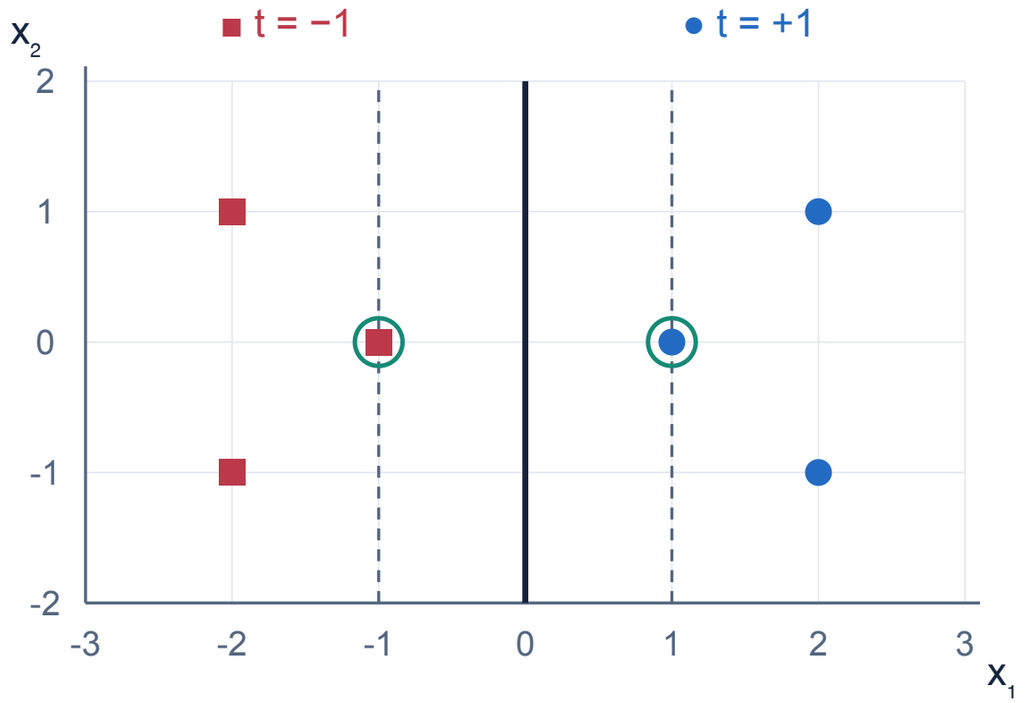
$$\min_i t_i(w^\top x_i + b) = 1$$

$$w^\top x + b = +1 \qquad \text{and} \qquad w^\top x + b = -1$$

Each margin plane is $1/\|w\|_2$ from the boundary.

> Full margin width: $2/\|w\|_2$.

Compute the width of the running example



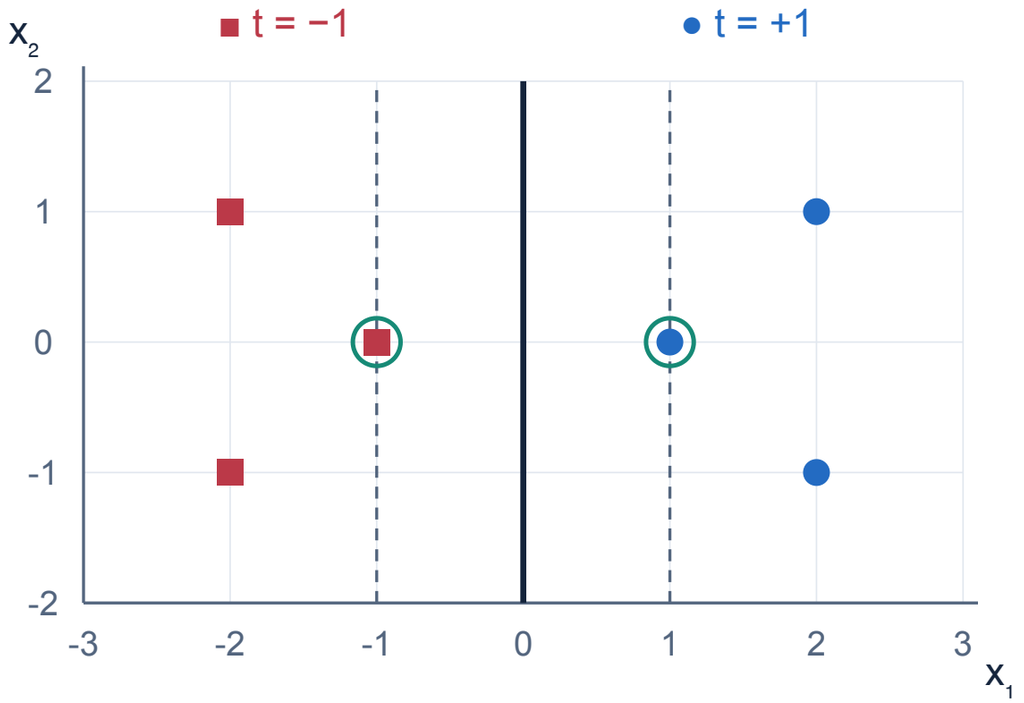
$w = (1, 0)^\top$, $b = 0$.

Margin planes: $x_1 = \pm 1$.

$$\|w\|_2 = 1, \quad \gamma = 1, \quad \text{width} = 2$$

Using $2w$ leaves the geometric boundary unchanged; its functional margin becomes 2.

Nearest constraints determine the separator



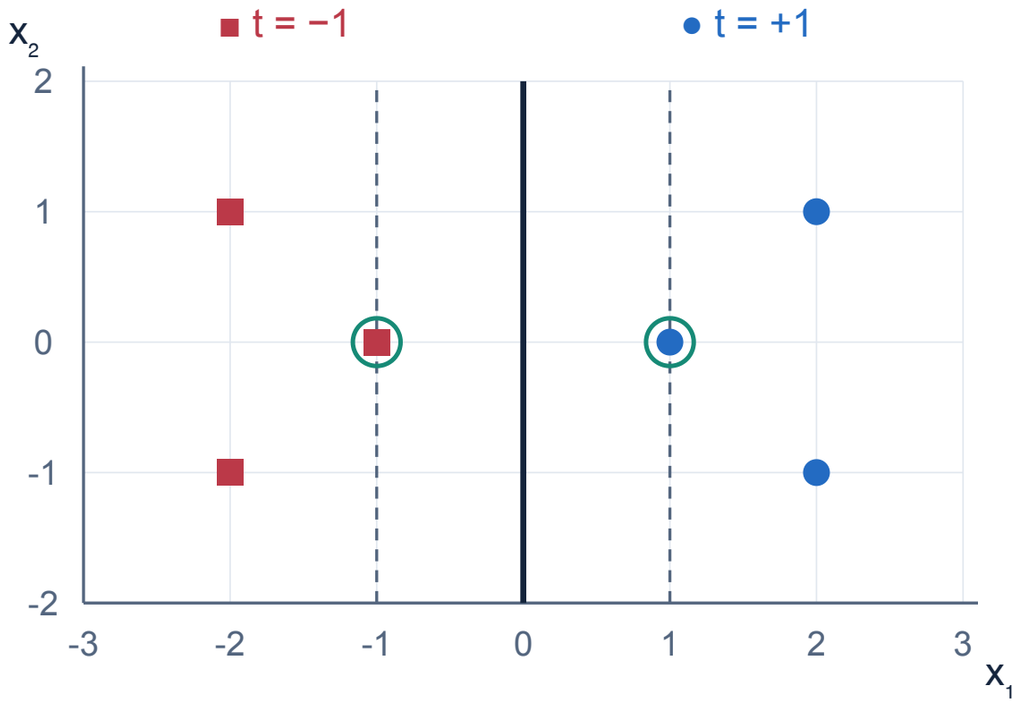
The closest examples touch the margin planes.

At the optimum, some training constraints are active.

> The dual assigns nonzero coefficients only to support vectors.

A touching point can have zero coefficient in a degenerate solution.

Rotate the boundary and measure its margin



Normal angle: **0°**

Intercept: **0.0**

The normal vector has unit length.

Minimum signed distance = 1.000

Which angle and intercept maximize the minimum signed distance?

Maximum margin becomes a convex problem

$$\min_{w,b} \ \frac{1}{2}\|w\|_2^2$$

$$\text{subject to} \quad t_i(w^\top x_i + b) \geq 1 \quad \text{for all } i$$

Minimizing the squared norm maximizes the canonical geometric margin.

> A convex quadratic objective with affine constraints: every local optimum is global.

Hard margin requires linearly separable data.

The dual problem

Replace a weight vector with one coefficient per training example.

Put the constraints into a Lagrangian

Write each constraint as $1 - t_i(w^\top x_i + b) \le 0$.

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2}\|w\|^2 + \sum_i \alpha_i [1 - t_i(w^\top x_i + b)]$$

$$\alpha_i \ge 0$$

> Each multiplier prices the corresponding training constraint.

Eliminate the primal variables

$$\nabla_w \mathcal{L} = w - \sum_i \alpha_i t_i x_i = 0$$

$$\boxed{w = \sum_i \alpha_i t_i x_i}$$

$$\frac{\partial \mathcal{L}}{\partial b} = -\sum_i \alpha_i t_i = 0 \quad \Longrightarrow \quad \sum_i \alpha_i t_i = 0$$

The equality constraint comes from the unregularized bias.

Substitute the weighted sum back in

Eliminate b using $\sum_i \alpha_i t_i = 0$:

$$\mathcal{L} = \frac{1}{2}\|w\|^2 + \sum_i \alpha_i - w^\top \sum_i \alpha_i t_i x_i$$

$$= \sum_i \alpha_i - \frac{1}{2}\left\|\sum_i \alpha_i t_i x_i\right\|^2$$

$$= \sum_i \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j t_i t_j \, x_i^\top x_j$$

Maximize a concave quadratic in α

$$\max_{\alpha} D(\alpha) = \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j t_i t_j \, x_i^\top x_j$$

$$\alpha_i \geq 0, \qquad \sum_i \alpha_i t_i = 0$$

> The data enter through inner products, not individual coordinates.

For separable data, strong duality gives the same optimum as the hard-margin primal.

Complementary slackness identifies support

$$\alpha_i [t_i (w^\top x_i + b) - 1] = 0$$

Outside the margin

$$t_i f(x_i) > 1 \implies \alpha_i = 0$$

No contribution to the weight sum.

Nonzero coefficient

$$\alpha_i > 0 \implies t_i f(x_i) = 1$$

The constraint is active.

Recover the classifier from support vectors

$$f(x) = \sum_{i \in \mathrm{SV}} \alpha_i t_i \, x_i^\top x + b$$

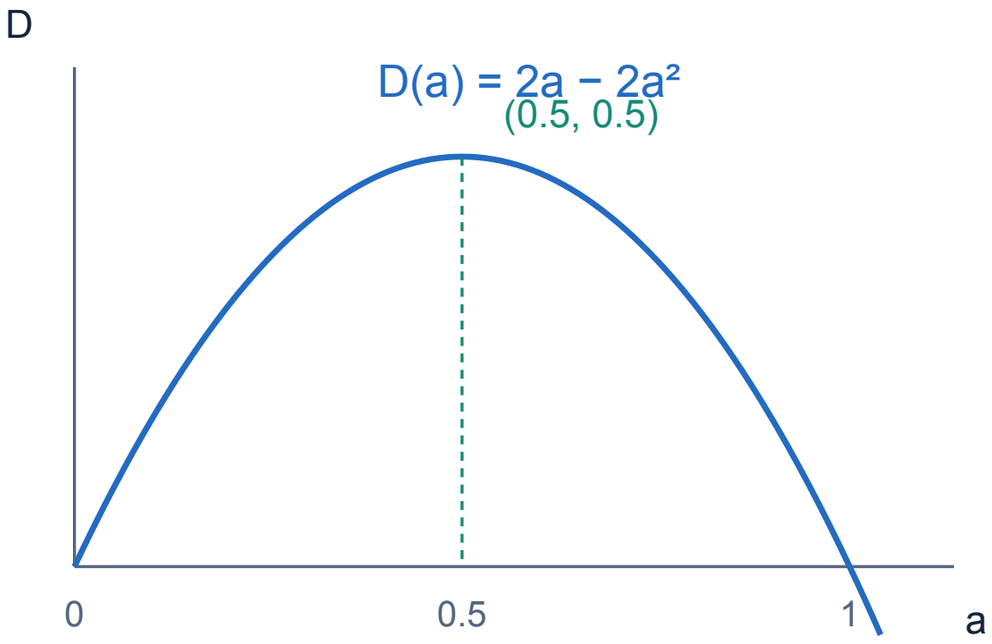
For a hard-margin support vector x_r:

$$b = t_r - \sum_{i} \alpha_i t_i \, x_i^\top x_r$$

> Train coefficients; predict using a weighted sum of similarities.

For a linear SVM, form and store w once.

Solve the dual for two support points



$x_- = (-1, 0),\ t_- = -1.$
$x_+ = (1, 0),\ t_+ = +1.$

The equality gives $\alpha_- = \alpha_+ = a$.

$$D(a) = 2a - 2a^2$$

$$a^* = \frac{1}{2}, \quad w = (1, 0)^\top, \quad b = 0$$

Primal = dual = 0.5.

Optimize two coefficients at a time

$$\sum_i \alpha_i t_i = 0$$

Changing one coefficient usually breaks the equality.

Move a pair along a feasible direction:

$$\alpha_i \leftarrow \alpha_i + t_i \delta, \qquad \alpha_j \leftarrow \alpha_j - t_j \delta$$

$$\delta^* = \operatorname{clip}_{[l,u]} \left(\frac{d^\top \nabla D(\alpha)}{d^\top Q d} \right)$$

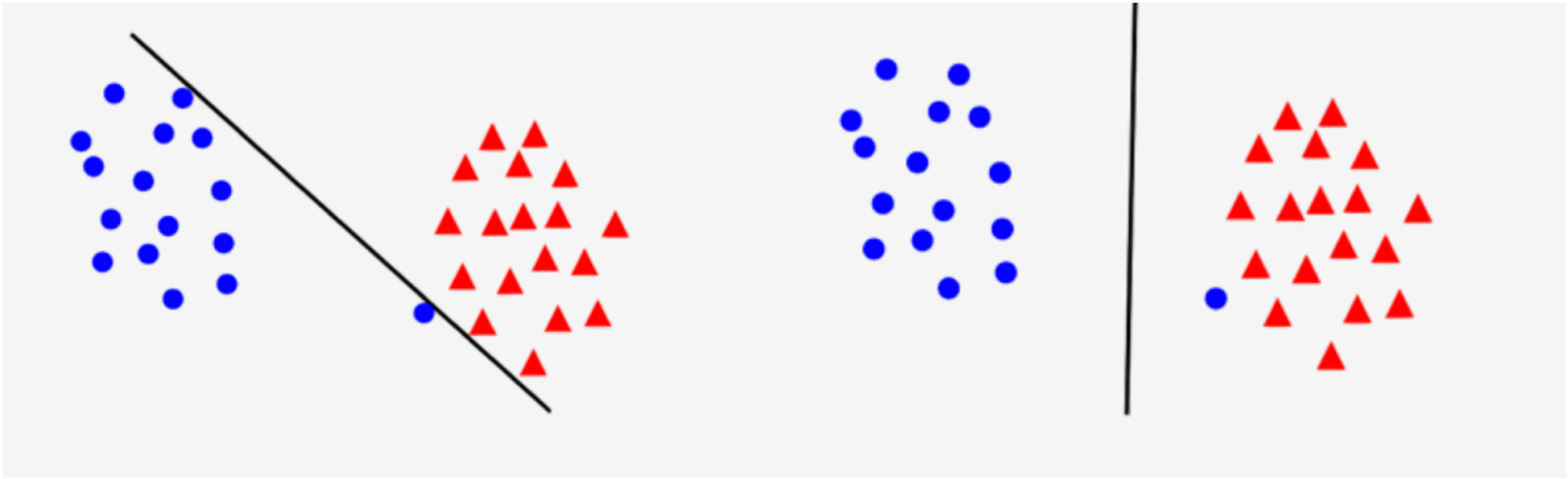
d is the pair direction; $Q_{ij} = t_i t_j x_i^\top x_j$. For positive curvature, clip to the feasible interval $[l, u]$.

Our tiny demos use exhaustive pair-coordinate ascent, not a production SMO implementation.

Allow controlled margin violations

Hard constraints can be brittle—or impossible—when examples overlap.

One unusual point can reshape the boundary

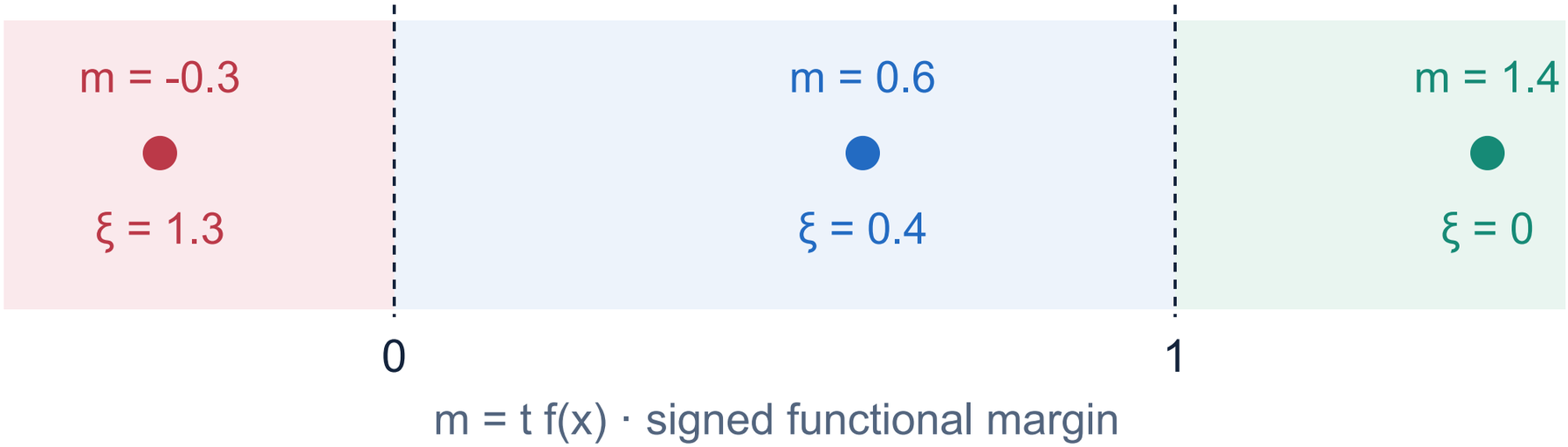


Insisting on zero training error may force a very narrow margin.

> Permit some violations, and charge for them explicitly.

Slack measures a shortfall from the margin

$$t_i(w^\top x_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0$$



The smallest feasible slack is $\xi_i = \max(0, 1 - t_i f(x_i))$.

A violation is not always a classification error

Signed score $m_i = t_i f(x_i)$	Minimum slack	Interpretation
$m_i \geq 1$	$\xi_i = 0$	On or beyond correct margin
$0 < m_i < 1$	$0 < \xi_i < 1$	Correct, inside the margin
$m_i = 0$	$\xi_i = 1$	On the decision boundary
$m_i < 0$	$\xi_i > 1$	Wrong side

Trade margin width against total slack

$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_i \xi_i$$

$$t_i(w^\top x_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0$$

Smaller C

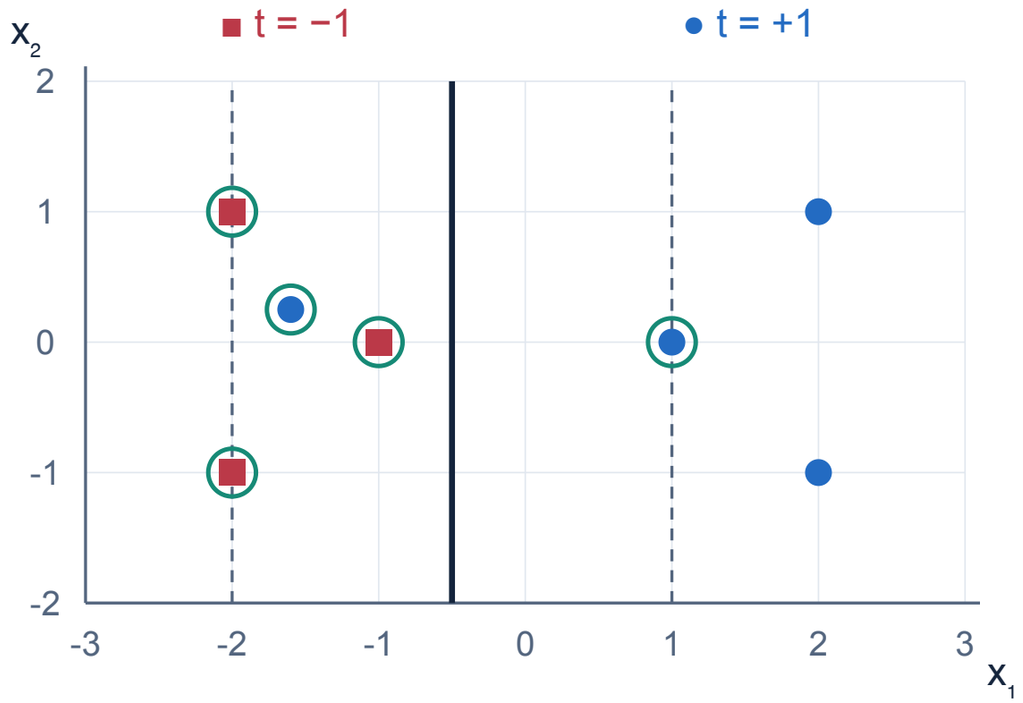
Violations cost less relative to the weight penalty.

Larger C

Violations cost more relative to the weight penalty.

Here $C > 0$ is a hyperparameter, not the number of classes.

Change C while keeping the data fixed



Slack penalty C

1

Same seven points, including an overlapping positive example.

Green circles: $\alpha_i > 0$.

1/7 training errors; 5 support vectors. Primal–dual gap: 1.4e-6

Computed linear SVM, synthetic data; no validation set.

Slack adds an upper bound in the dual

Add multipliers $\mu_i \geq 0$ for $-\xi_i \leq 0$.

$$\frac{\partial \mathcal{L}}{\partial \xi_i} = C - \alpha_i - \mu_i = 0$$

$$\boxed{0 \leq \alpha_i \leq C}$$

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j t_i t_j x_i^\top x_j$$

$$\text{subject to } 0 \leq \alpha_i \leq C, \quad \sum_i \alpha_i t_i = 0$$

Read support-vector roles from α

Coefficient at optimum	Signed score	Role
$\alpha_i = 0$	$t_i f(x_i) \geq 1$	No contribution to prediction
$0 < \alpha_i < C$	$t_i f(x_i) = 1$	Free margin support vector
$\alpha_i = C$	$t_i f(x_i) \leq 1$	On or inside margin

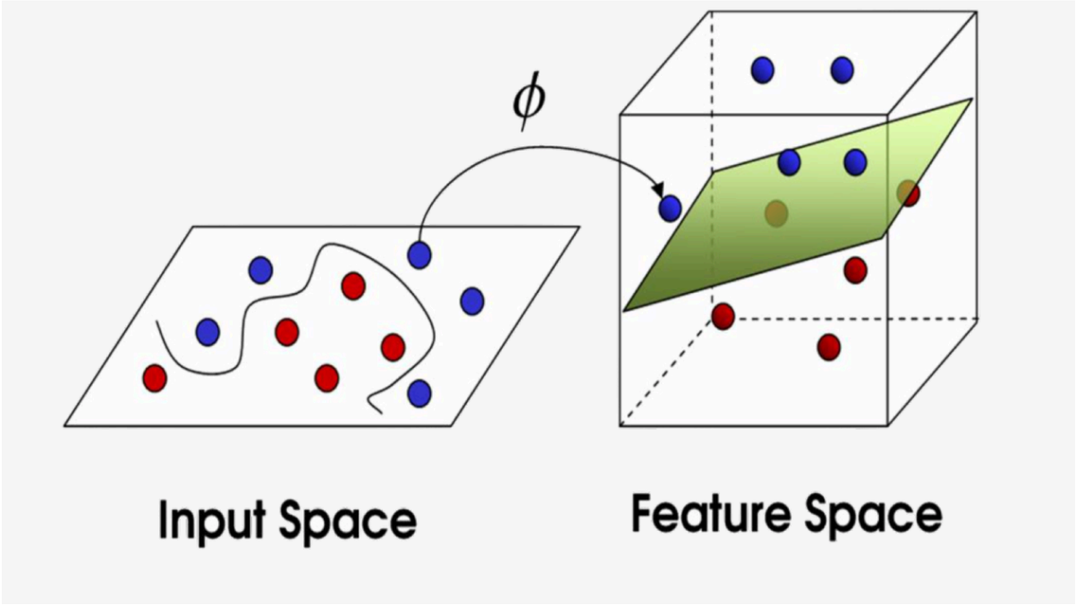
> Recover $b = t_i - \sum_j \alpha_j t_j k(x_j, x_i)$ from a free margin support vector.

With no free support vector, use the KKT bounds on b.

Nonlinear boundaries through kernels

The dual needs similarities between examples.
Can we compute them in a richer feature space?

Separation can appear after a feature mapping

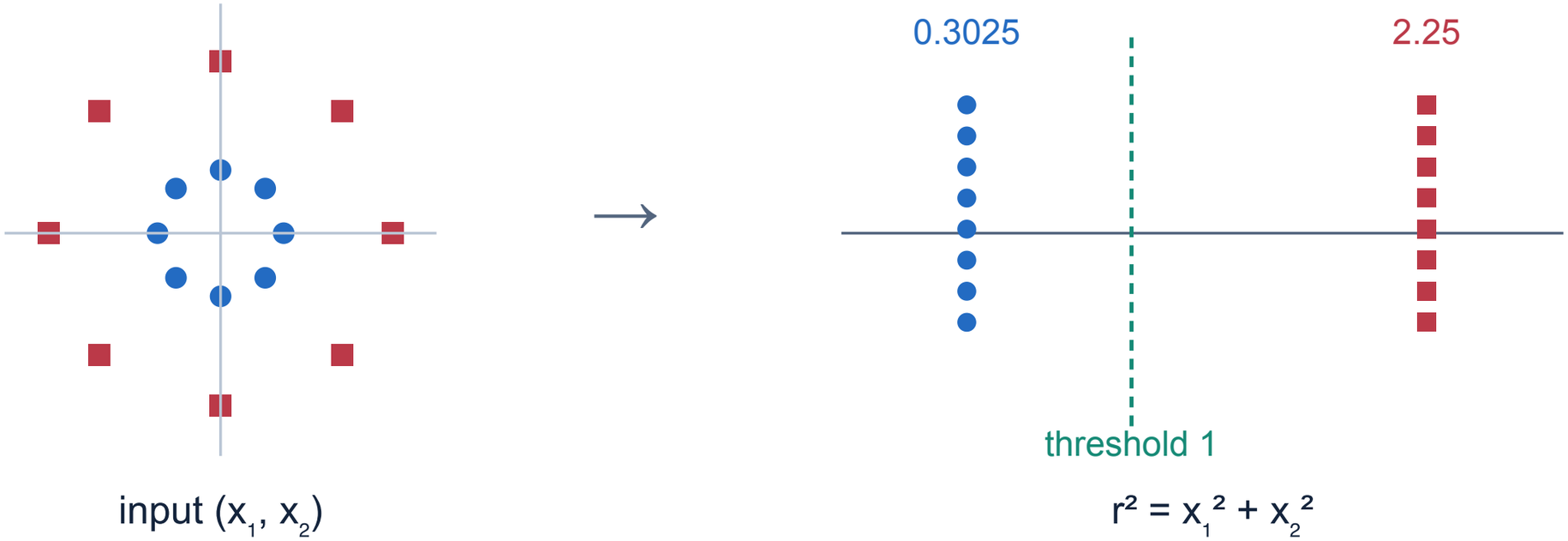


Choose a feature map $\phi(x)$.

$$f(x) = w^\top \phi(x) + b$$

A hyperplane in feature space can be curved in input coordinates.

A nonlinear feature separates the rings



$$\phi(x) = x_1^2 + x_2^2, \qquad f(x) = 1 - \phi(x)$$

Evaluate inner products without the features

$$k(x, z) = \phi(x)^\top \phi(z)$$

The dual objective and prediction only need these values.

$$x_i^\top x_j \quad \longrightarrow \quad k(x_i, x_j)$$

> Compute a feature-space inner product without constructing each feature vector.

The feature map is implicit; an optimization problem still has to be solved.

Expand a kernel once to see why it works

$$k(x, z) = (1 + x^\top z)^2$$

For $x = (x_1, x_2)$, one feature map is

$$\phi(x) = \left(1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, \sqrt{2}x_1x_2, x_2^2\right)^\top$$

$$\phi(x)^\top \phi(z) = (1 + x_1z_1 + x_2z_2)^2$$

> $x = (1, 2)$, $z = (3, -1)$: $k(x, z) = (1 + 1)^2 = 4$.

The dual now gives a nonlinear classifier

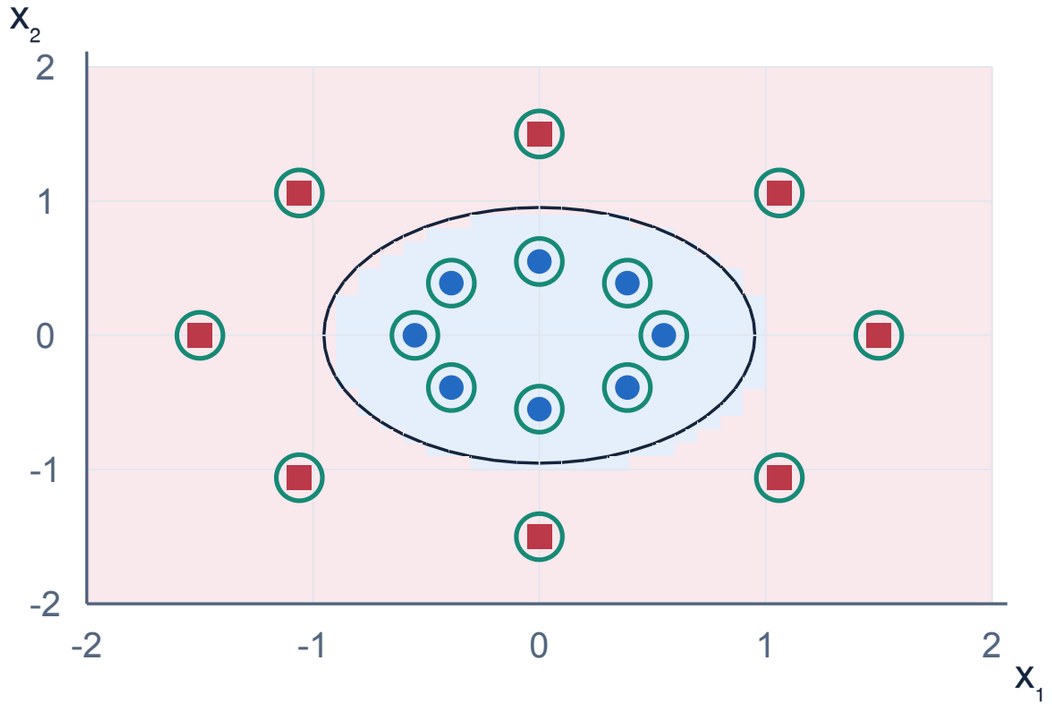
$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j t_i t_j k(x_i, x_j)$$

$$0 \le \alpha_i \le C, \qquad \sum_i \alpha_i t_i = 0$$

$$f(x) = \sum_{i \in \mathrm{SV}} \alpha_i t_i k(x_i, x) + b$$

> The optimizer still learns coefficients on training examples.

RBF bandwidth changes the shape of influence



$$k(x, z) = e^{-\gamma \|x - z\|_2^2}$$

RBF parameter γ

1

$C = 10$; same 16 ring points.

16 support vectors. Primal–dual gap: 1.9e-5

Small γ: broad influence.
Large γ: local influence.

A valid kernel behaves like an inner product

For every finite input set, the Gram matrix $G_{ij} = k(x_i, x_j)$ must be symmetric and positive semidefinite.

$$v^\top G v \geq 0 \qquad \text{for every real vector } v$$

Linear	$k(x, z) = x^\top z$
Polynomial	$k(x, z) = (x^\top z + c)^d$, $c \geq 0$, positive integer d
RBF	$k(x, z) = e^{-\gamma \lVert x - z \rVert^2}$, $\gamma > 0$

Fit scaling on training data; select kernel settings and C on validation data.

Choose the optimization view for the problem

	Explicit linear features	Implicit nonlinear kernel
Parameters	Store w, b	Support vectors, coefficients, bias
Typical route	Primal or linear dual solvers	Kernel dual / SMO-like methods
Prediction	One dot product	One kernel per support vector

A full dense Gram matrix has N^2 entries; solvers can cache subsets.

> Kernels enrich fixed features. An MLP learns its representation jointly with the classifier.

From two classes to many

Keep the margin idea.

Decide how to compare more than two class scores.

Combine binary classifiers—or train jointly

One versus rest

Train K binary classifiers.

Choose the largest class-versus-rest score.

One versus one

Train $K(K-1)/2$ binary classifiers.

Use pairwise votes with a tie rule.

> These reductions differ from optimizing a joint multiclass loss.

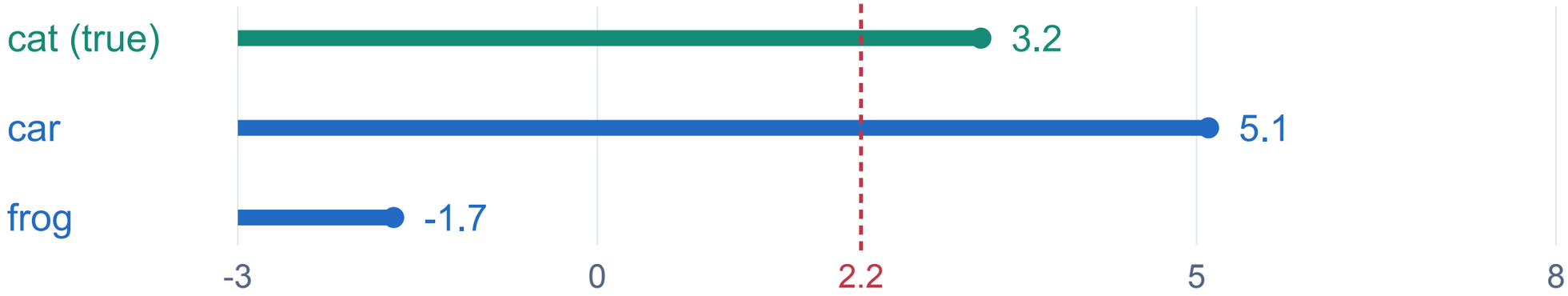
Three classes give 3 one-vs-rest models or 3 pairwise models.

A matrix produces one score per class

$$s = Wx + b, \qquad W \in \mathbb{R}^{K \times D}, \quad b \in \mathbb{R}^{K}$$

$$\hat{y} = \underset{j \in \{1, \ldots, K\}}{\arg\max}\ s_j$$

True label y_i is now a class index, not a signed label.



The true class must beat every competitor

$$s_{y_i} \geq s_j + 1 \quad \text{for every } j \neq y_i$$

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

> Add one hinge penalty for each violated class-score margin.

Exclude the true class. This is CS231n’s sum-of-hinges formulation.

Calculate the loss before changing the model

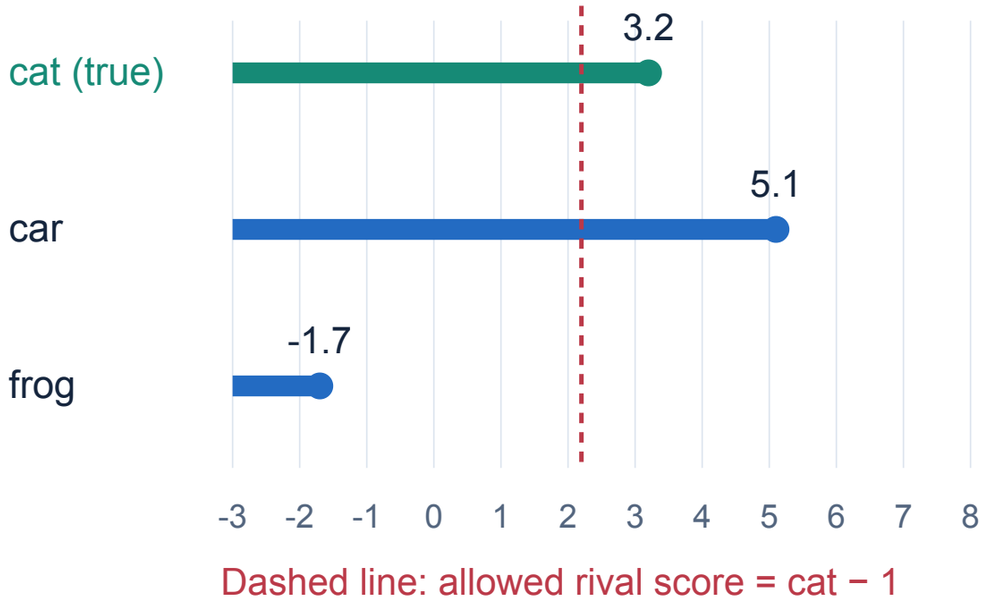
Class	Cat image	Car image	Frog image
cat	3.2	1.3	2.2
car	5.1	4.9	2.5
frog	−1.7	2.0	−3.1

$$L_{\text{cat}} = \max(0, 5.1 - 3.2 + 1) + \max(0, -1.7 - 3.2 + 1) = 2.9$$

Car: 0. Frog: $6.3 + 6.6 = 12.9$.

Mean data loss: $(2.9 + 0 + 12.9)/3 = 5.2667$.

Correct predictions can still violate a margin



True cat score: **3.2**

Car score: **5.1**

Frog score fixed at −1.7.

Hinge loss = 2.90

Cat 3.2 and car 3.0: prediction is correct. Is the loss zero?

Eliminating slack exposes the binary hinge

$$\xi_i^* = \max(0, 1 - t_i(w^\top x_i + b))$$

$$\min_{w,b} \ \frac{1}{2}\|w\|^2 + C\sum_i \max(0, 1 - t_i f(x_i))$$

> The constrained soft-margin problem and regularized hinge-loss problem have the same minimizers.

Match the sum and mean conventions

$$\frac{1}{2}\|w\|^2 + C\sum_i \ell_i$$

Divide by the positive constant CN:

$$\frac{1}{N}\sum_i \ell_i + \frac{\lambda}{2}\|w\|^2, \qquad \lambda = \frac{1}{CN}$$

> For this fixed dataset and these conventions, larger C means smaller λ.

Check normalization: other objectives may average differently or omit the half.

A joint loss gives a direct learning problem

$$L(W, b) = \frac{1}{N} \sum_i \sum_{j \neq y_i} \max(0, s_{ij} - s_{i,y_i} + 1) + \frac{\lambda}{2} \|W\|_F^2$$

$s_i = W x_i + b$; the bias is not regularized.

The joint objective is convex for linear scores.

> Optimize the weights directly without deriving another kernel dual.

A violated margin tells scores how to move

Let $V_i = \{j \neq y_i : s_{ij} - s_{i,y_i} + 1 > 0\}$.

$$\frac{\partial L_i}{\partial s_{ij}} = 1\,(j \in V_i), \qquad \frac{\partial L_i}{\partial s_{i,y_i}} = -|V_i|$$

Other competing scores have derivative zero.

Default cat example: only car violates.

$$\nabla_s L_i = (-1, 1, 0)^\top$$

> A descent step raises the true score and lowers the violating score.

At a hinge kink, choose a valid subgradient; zero is a common convention.

Convert score gradients into weight gradients

For one example, set $d_i = \nabla_{s_i} L_i$.

$$\nabla_W L_i = d_i x_i^\top, \qquad \nabla_b L_i = d_i$$

$$g_W = \frac{1}{B} \sum_{i \in \mathcal{B}} d_i x_i^\top + \lambda W, \qquad g_b = \frac{1}{B} \sum_{i \in \mathcal{B}} d_i$$

$$W \leftarrow W - \eta g_W, \qquad b \leftarrow b - \eta g_b$$

Learning rate $\eta > 0$; B is the actual minibatch size.

Work one binary subgradient update

$x = (1, 2)^\top$, $t = +1$, $w = (0.2, -0.1)^\top$, $b = 0$.

Use $\lambda = 0.1$, $\eta = 0.1$. Here $tf(x) = 0 < 1$.

$$g_w = \lambda w - tx = (-0.98, -2.01)^\top, \qquad g_b = -1$$

$$w^+ = (0.298, 0.101)^\top, \qquad b^+ = 0.1$$

New signed score: $0.298 + 2(0.101) + 0.1 = 0.6$.
Hinge data loss: $1 \to 0.4$.

A sampled update need not reduce the full dataset loss.

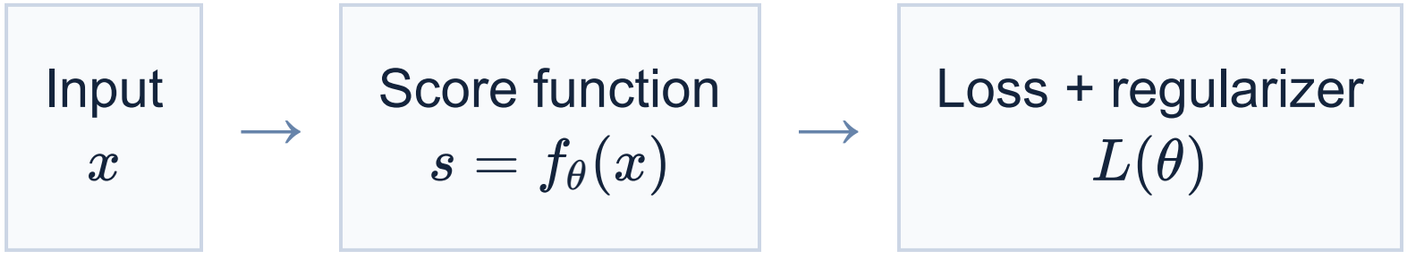
A minibatch subgradient method scales training

```
# Batch rows; use the current weights to form all gradients.
for Xb, yb in shuffled_minibatches(X, y):
    scores = Xb @ W.T + b
    d = multiclass_hinge_score_grad(scores, yb)
    W -= eta * (d.T @ Xb / len(Xb) + lam * W)
    b -= eta * d.mean(axis=0)
```

Repeat: sample → score → measure violations → update.

Code labels 0…K−1. The helper returns score gradients for each example’s summed hinge loss.

Keep the objective; change the score function



Linear scores

$$f_\theta(x) = Wx + b$$

Learned features

$$f_\theta(x) = W_2\phi(W_1x + b_1) + b_2$$

> Compute a loss, differentiate, and update parameters.

Connect geometry and the learning view

1. **Margin:** normalize scores to compare distances.
2. **Dual:** optimize coefficients on examples; KKT reveals support.
3. **Soft margin:** trade violations against weight size.
4. **Kernel:** compute feature-space inner products implicitly.
5. **Multiclass:** compare scores, sum hinge losses, optimize weights.

Check the connections

1. Canonical $\|w\| = 2$: what is the full margin width?
2. Signed score $tf(x) = 0.6$: what is the minimum slack? Is the prediction correct?
3. True score2, rivals1.5 and−1: what is the summed multiclass hinge loss?
4. Why does the dual enable the kernel trick?

1. 1. 2. 0.4, correct. 3. 0.5.
4. It needs only pairwise inner products.

Reading and source lectures

- **Harvard CS109A · Lecture 20**

  Geometry, hard/soft margins, feature-space intuition.

- **Stanford CS229 · §§6.3–6.8 and Chapter 5**

  Margins, duality, SMO, and valid kernels.

- **CS231n · Linear Classification · Optimization I**

  Worked multiclass losses and the learning perspective.

- **Platt · SMO · Chang & Lin · LIBSVM**

  Supplementary: practical optimization and multiclass reductions.